

Data Appendix: Sea Ice Data

Dataset 1: Cleaned Sea Ice Data

This is our only dataset we used. The unit of observation is in **millions of sq km**

- In **sea_ice_data_clean.csv**
- Used to train a linear regression model, a quadratic model, and a ARIMA model

Variables

564 observations and 5 columns

- When: **year** and **mo**
- Data Source of Observation: **source_dataset** (ex. NSIDC-0051, NSDIC-0803)
- Observations:
 - **extent**: the sea ice extent
 - **area**: the area of the sea ice

Figures & Outputs

Figure 1 (Sea Ice Extent - Time Series): This shows the observed measurements of the sea ice extent over time.

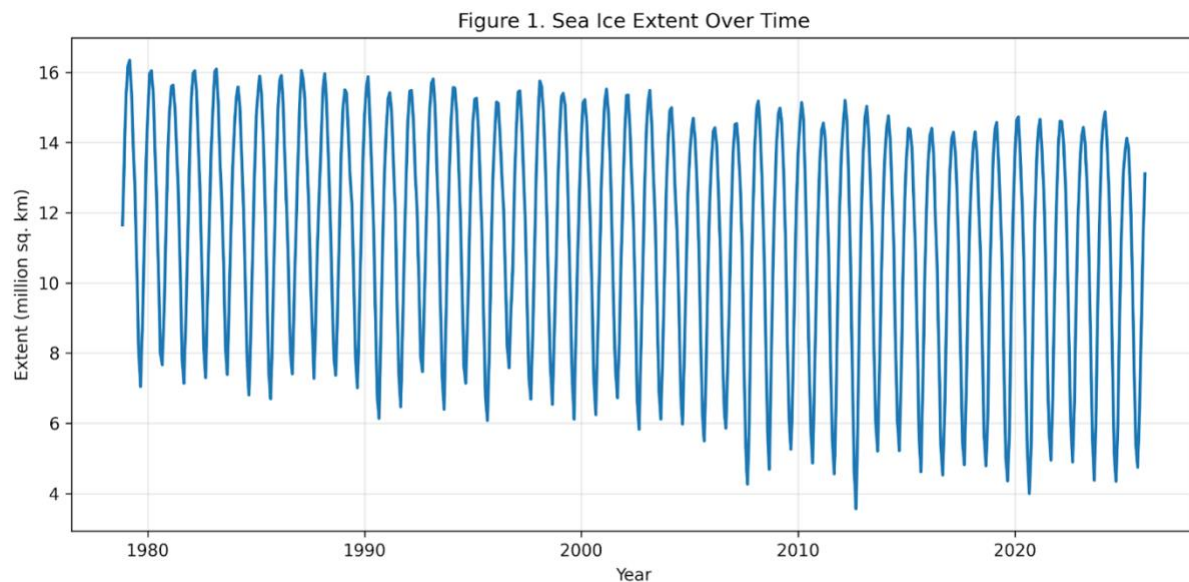


Figure 2 (Sea Ice Area - Time Series): This shows the observed measurements of the sea ice area over time.

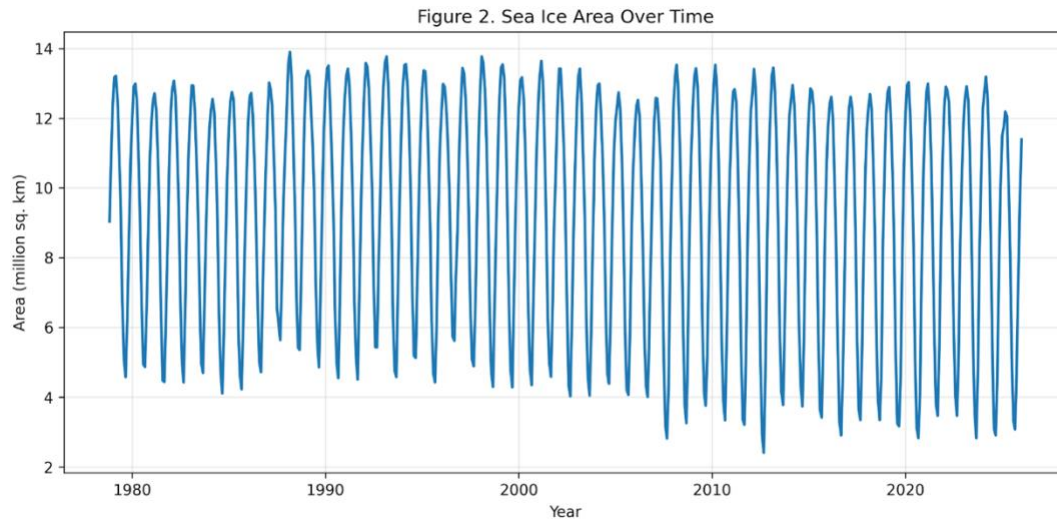


Figure 3 (Average Sea Ice Extent by Month): Represents the average extent of sea ice in each month over the time frame.

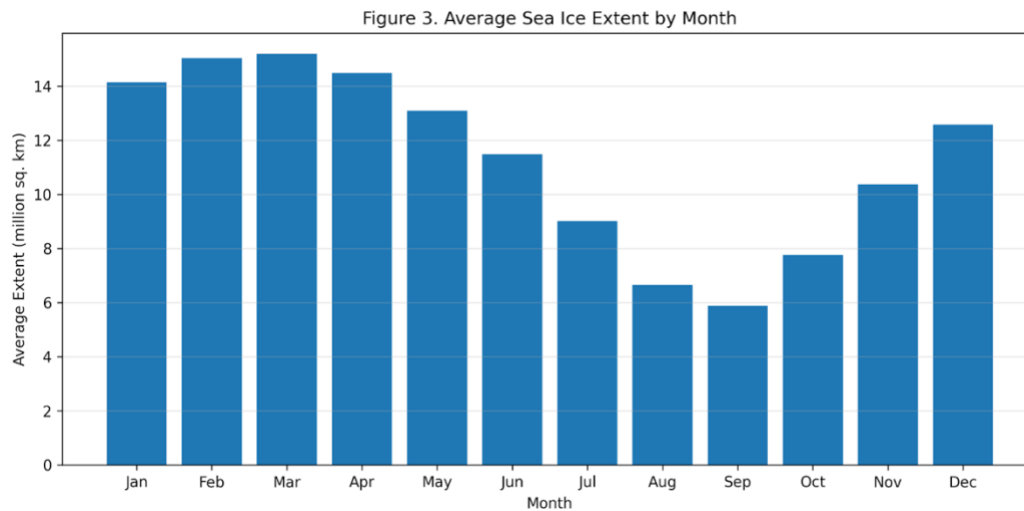


Figure 4 (Spring vs. Fall Ice Extent OVER Time): Represents the sea ice extent over time in the spring and fall seasons

Figure 4. March vs September Sea Ice Extent Over Time

